

Measuring the Topology of Word Embeddings

Benjamin Matthias Ruppik

Dialog Systems and Machine Learning Lab,
Faculty of Mathematics and Natural Sciences,
Heinrich Heine University Düsseldorf, Germany

Plan

What

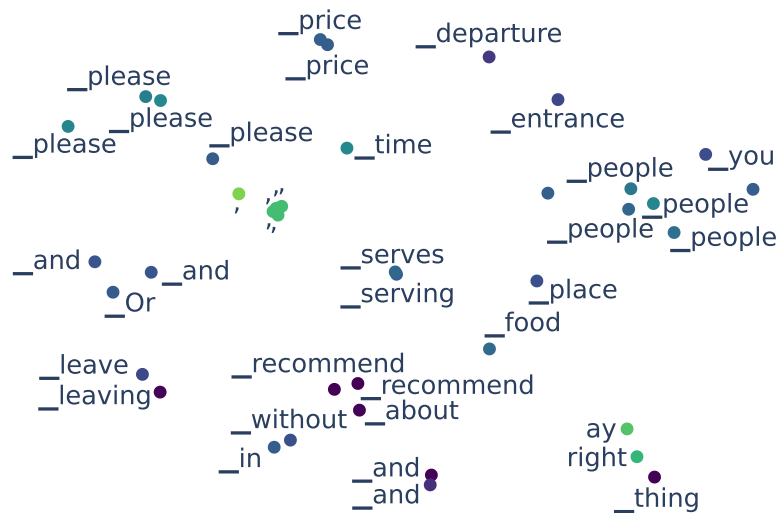
**Word Embeddings &
Latent Spaces of LLMs**

How

Local Intrinsic Dimension

Why

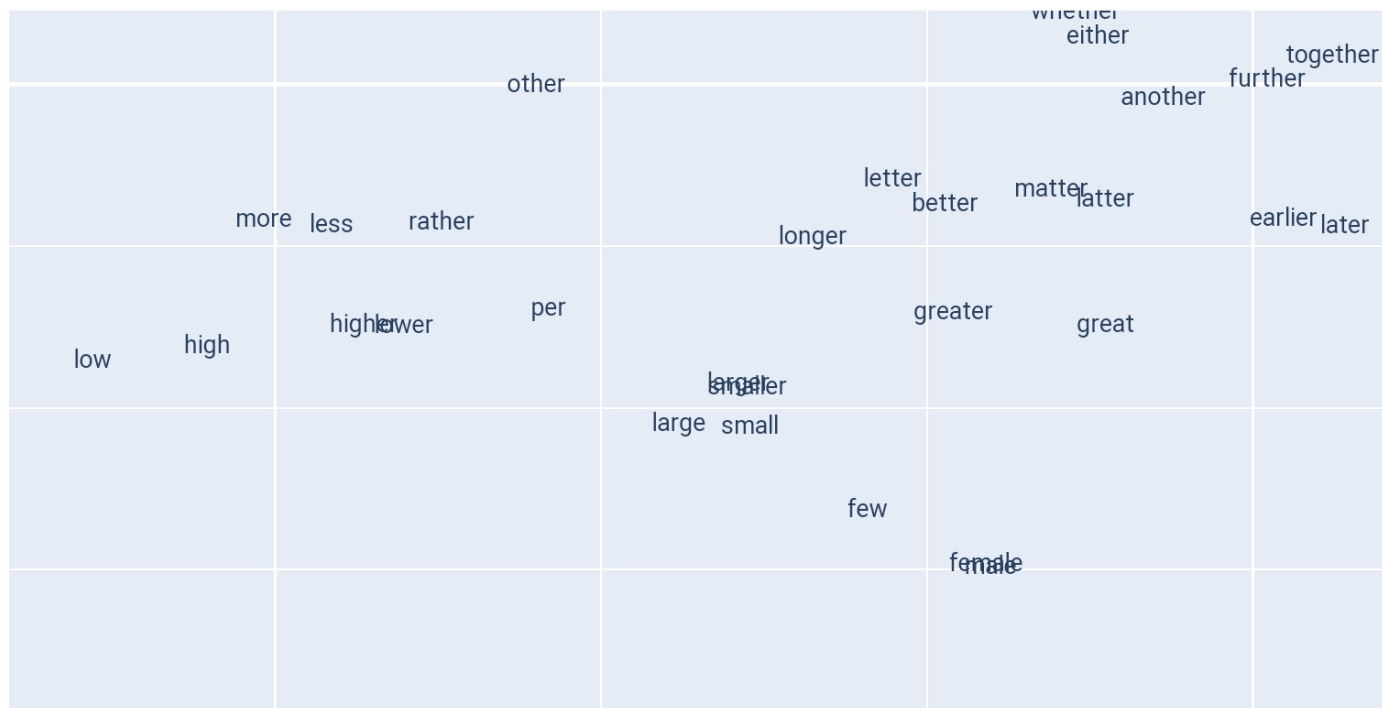
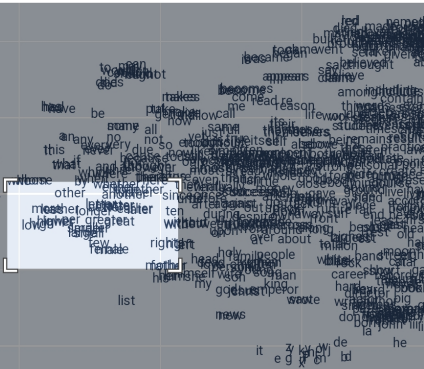
**Using Local Intrinsic Dimension
to Monitor Training Dynamics**



- “I need to book a cab for 1 person to get there. I would be alright with a shared ride.”
- “You want to play Be Alright on the TV, is that correct?”

- “I need to book a cab for 1 person to get there. I would be alright with a shared ride.”
- “You want to play Be Alright on the TV, is that correct?”

- “I need to book a **cab** for **1** person to get there. I would be alright with a **shared ride**.”
- “You want to play Be Alright on the **TV**, is that correct?”



What does recent English borrowing *ongchoi* mean?

Suppose you see these sentences:

- Ong choi is delicious **sautéed with garlic**.
- Ong choi is superb **over rice**
- Ong choi **leaves** with salty sauces

And you've also seen these:

- ...spinach **sautéed with garlic over rice**
- Chard stems and **leaves** are **delicious**
- Collard greens and other **salty** leafy greens

Conclusion:

- Ongchoi is a leafy green like spinach, chard, or collard greens
- We could conclude this based on words like "leaves" and "delicious" and "sauteed"

Ongchoi: *Ipomoea aquatica* "Water Spinach"

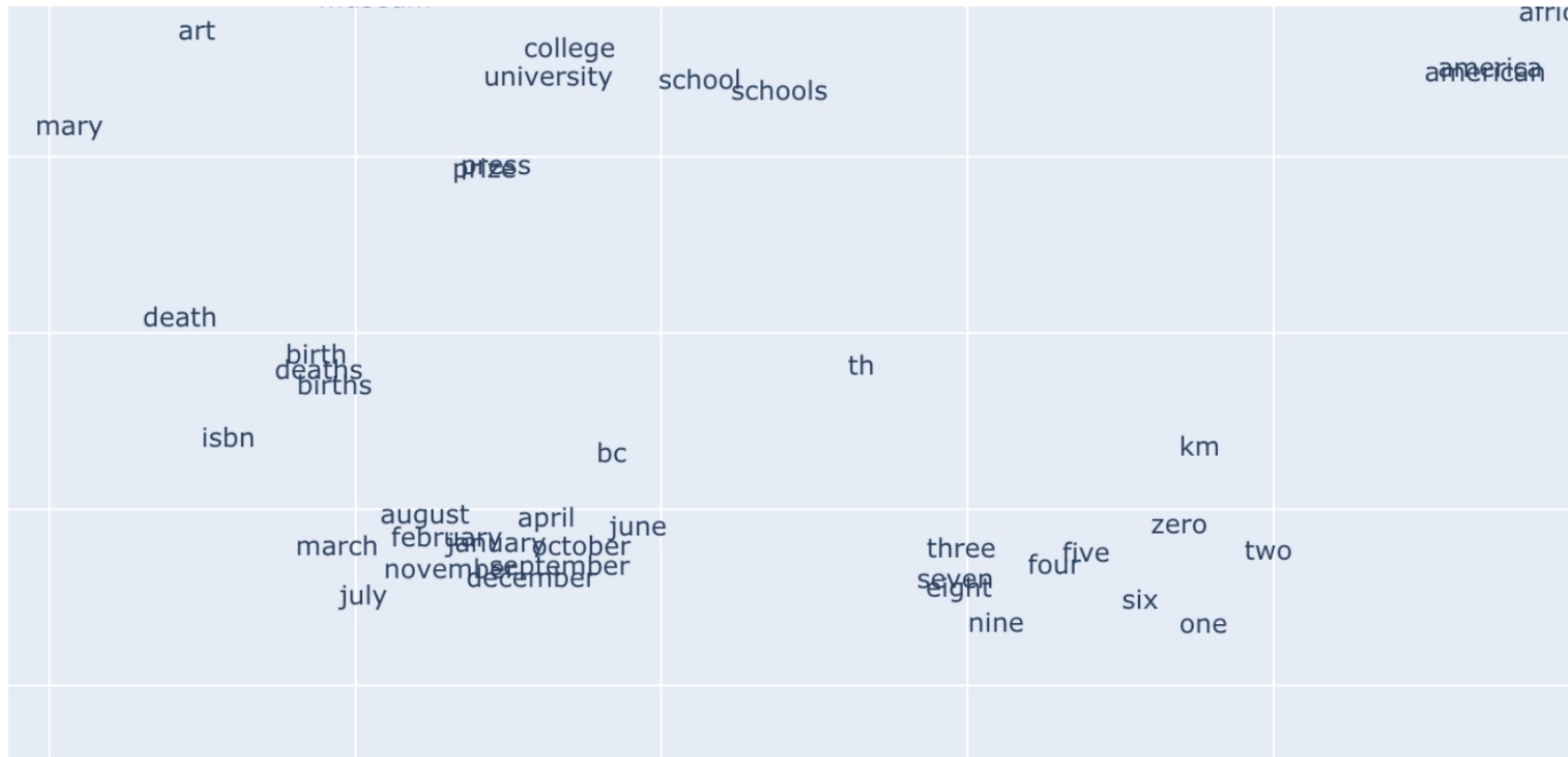
空心菜
kangkong
rau muống
...



Yamaguchi, Wikimedia Commons, public domain

Slides taken from [Jurafsky, Martin] Speech and Language Processing (2025), <https://web.stanford.edu/~jurafsky/slp3/>

Word Embeddings



Tokenization

- Language models process text in chunks called **tokens**
- Tokens represent commonly occurring sequences of characters
- **Example:**
“Tekniska” decomposed as
“Tek” and “nis” and “ka”
- Short and common words like “the” represented as single token

GPT-4o & GPT-4o miniGPT-3.5 & GPT-4GPT-3 (Legacy)

KTH Royal Institute of Technology (Swedish: Kungliga Tekniska högskolan, lit. 'Royal Technical High School'), abbreviated KTH, is a public research university in Stockholm, Sweden.

ClearShow example

Tokens

41

Characters

180

KTH Royal Institute of Technology (Swedish: Kungliga Tekniska högskolan, lit. 'Royal Technical High School'), abbreviated KTH, is a public research university in Stockholm, Sweden.

<https://platform.openai.com/tokenizer>

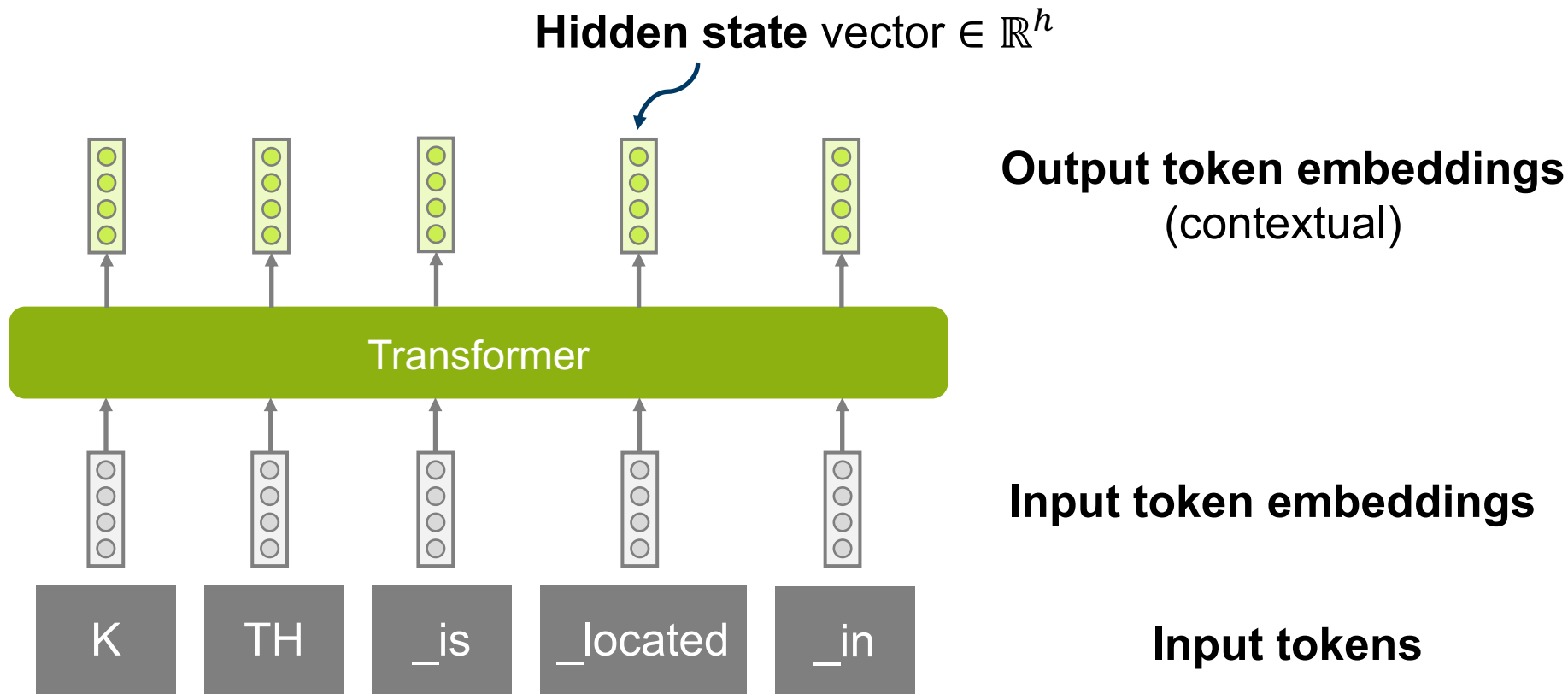
Tokenization

KTH Royal Institute of Technology (Swedish: Kungliga Tekniska högskolan, lit. 'Royal Technical High School'), abbreviated KTH, is a public research university in Stockholm, Sweden.

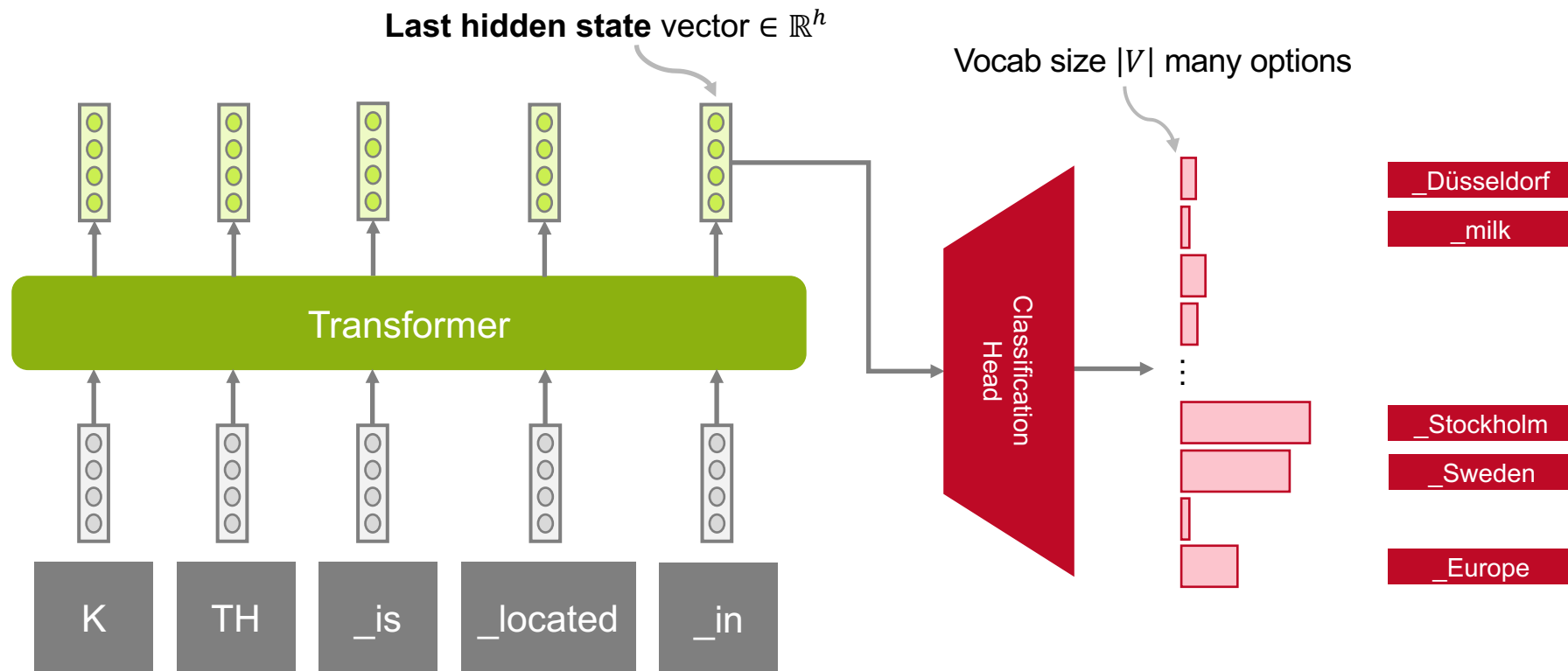
KTH Royal Institute of Technology (Swedish: Kungliga Tekniska högskolan, lit. 'Royal Technical High School'), abbreviated KTH, is a public research university in Stockholm, Sweden.

[42, 4867, 18516, 14544, 328, 15101, 350, 15254, 45708, 25, 57124, 28086, 39872, 12113, 1854, 79450, 3817, 47537, 11, 11980, 13, 461, 79403, 35605, 6597, 6586, 10047, 179588, 658, 4867, 11, 382, 261, 751, 4176, 16490, 306, 60611, 11, 42009, 13]

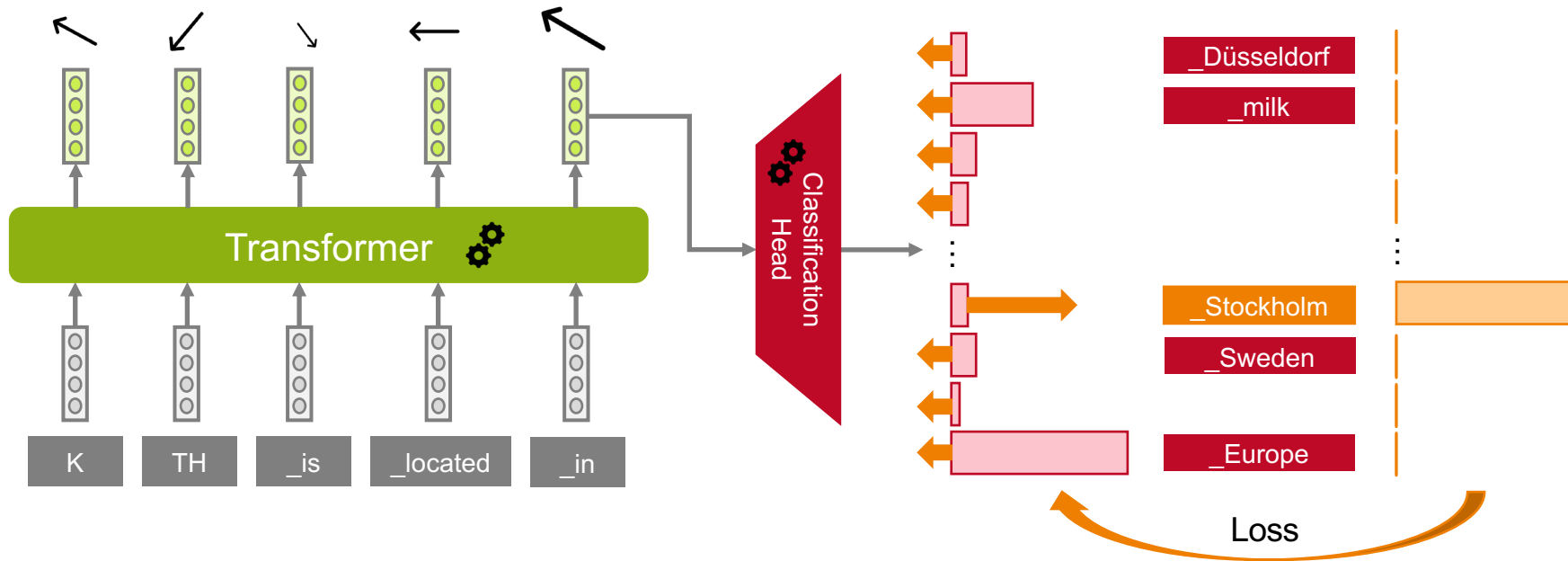
Contextual Language Models



Inference: Predicting the Next Token



Unsupervised Pre-Training



Training Example:

K

TH

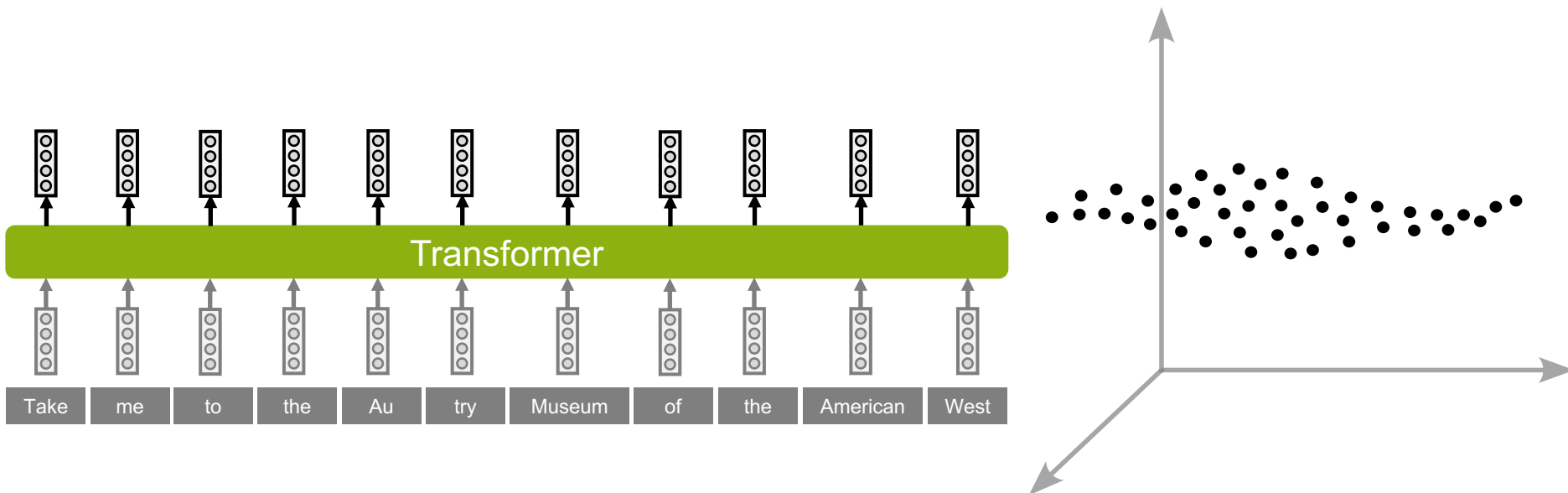
_is

_located

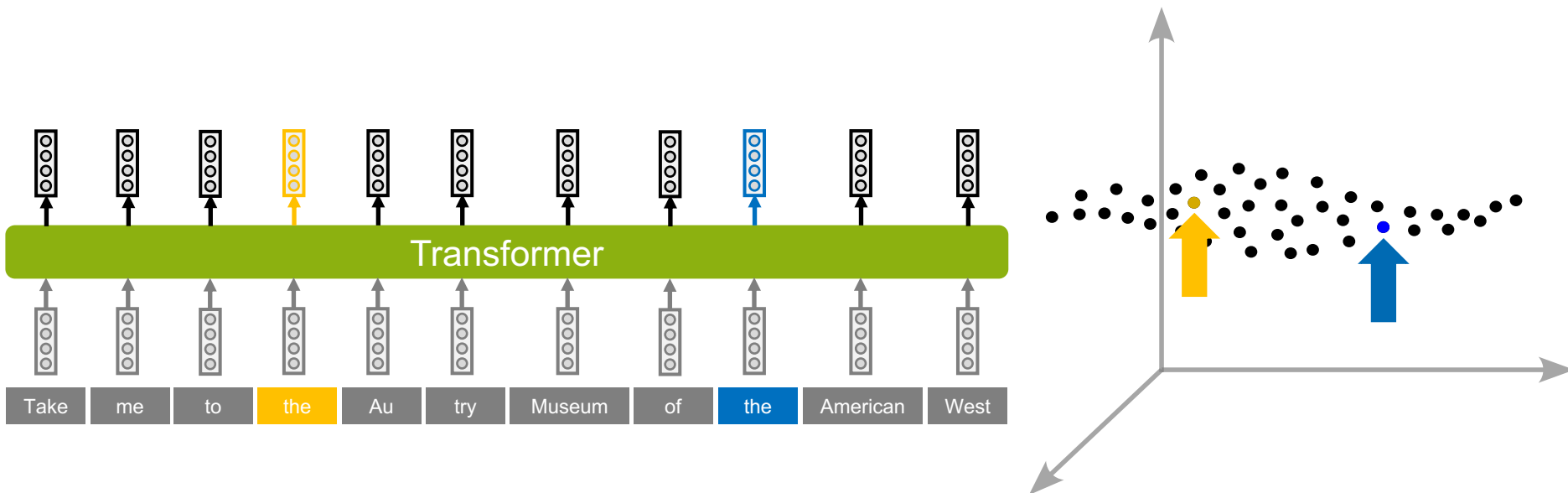
_in

_Stockholm

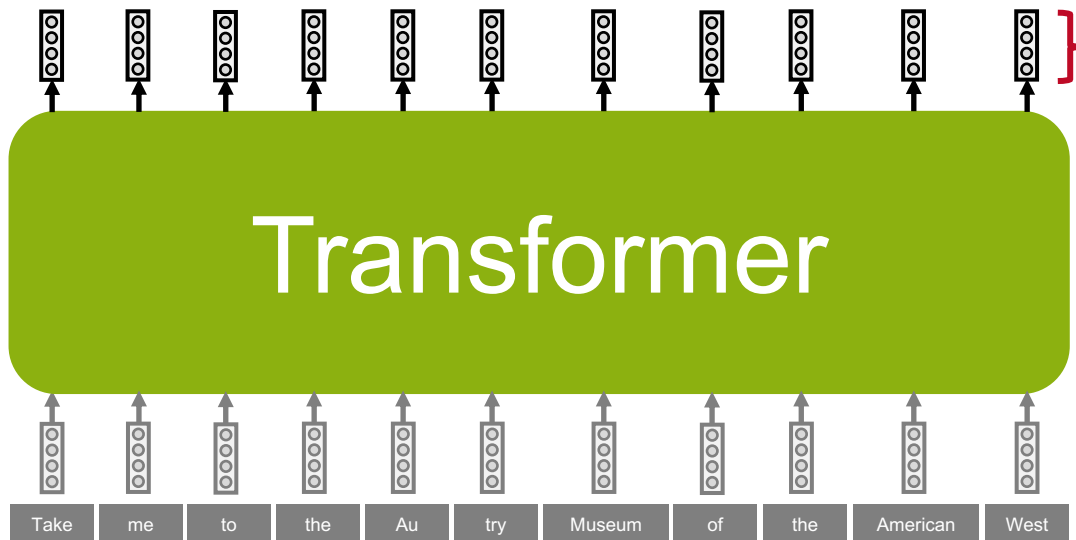
Contextual Language Models



Contextual Language Models



Modern LLMs are literally **Large**



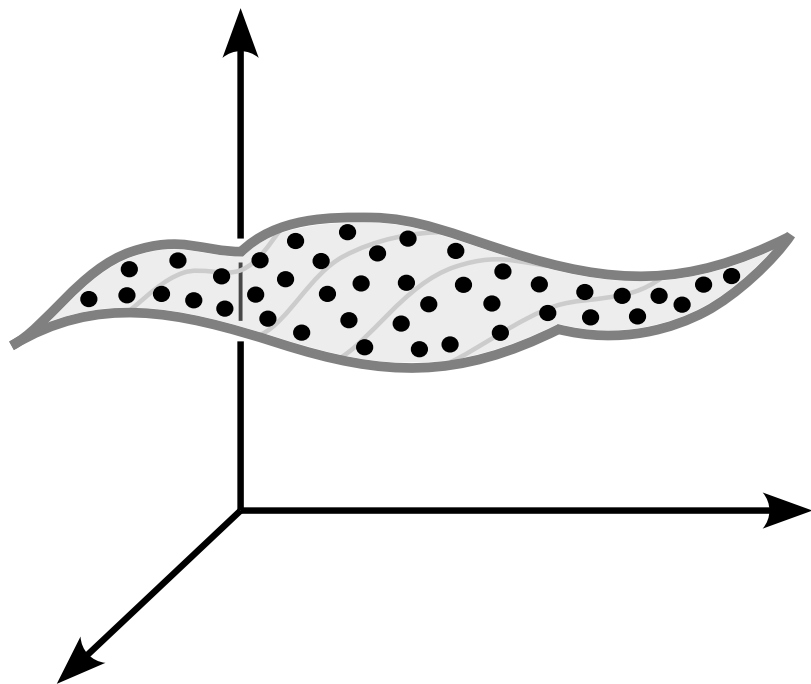
Example: Llama-3.1-8B

- **Hidden Size: 4,096**
- Vocabulary Size: 128,256
- Download Size: 32.1 GB

Example: Llama-3.1-70B

- **Hidden Size: 8,192**
- Vocabulary Size: 128,256
- Download Size: 282 GB

Manifold Hypothesis

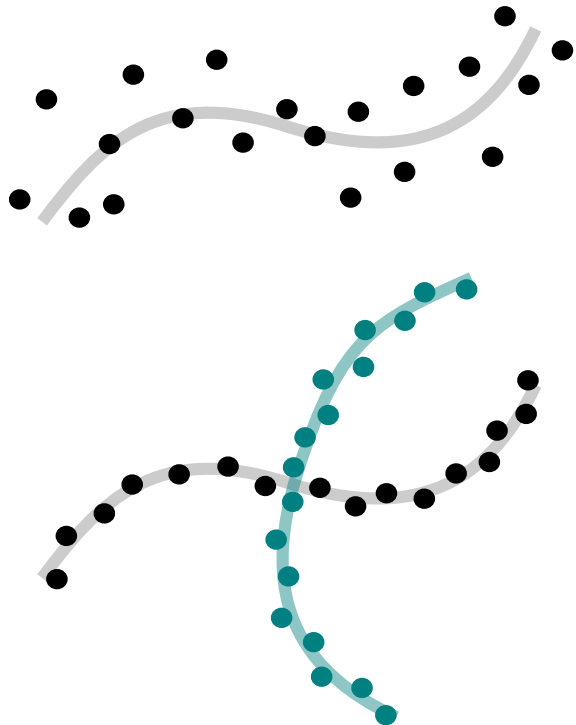


High-dimensional data lies on or near a union of **low-dimensional manifold structures**.

Consequences:

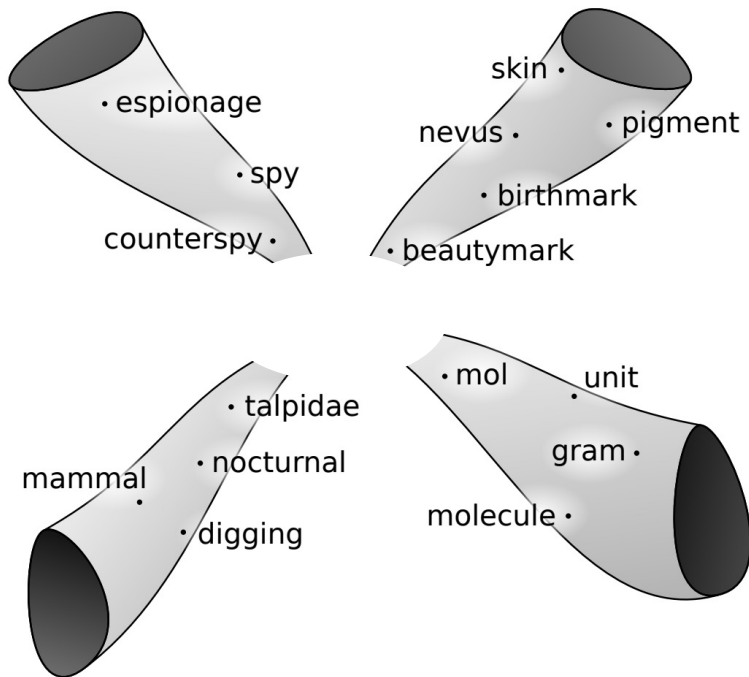
- Most variation in the data **captured by small number** of dimensions
- Explains effectiveness of **dimensionality reduction** techniques (PCA, t-SNE, UMAP, ...)

Problems with Manifold Hypothesis



- **Noise**
- **Limited number of samples:**
impossible to accurately estimate the underlying structure
- **Union of manifolds:** If data has multiple underlying structures → difficult to map onto a single reduced space
- **Singularities**

Singularities in Static Word Embedding Spaces



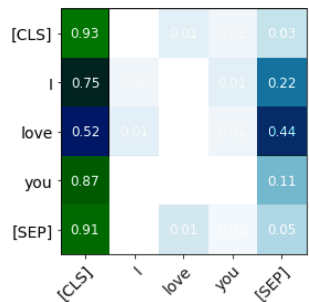
Polysemous word:
Word with multiple meanings

Number of meanings of a word:

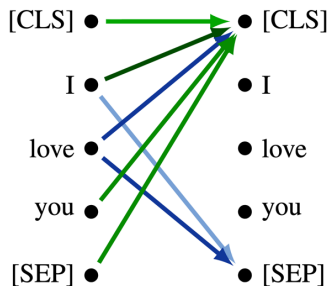
Reflected by the **number of components of a punctured neighborhood** of a word

Alexander Jakubowski, Milica Gasic, and Marcus Zibrowius. 2020. Topology of Word Embeddings: Singularities Reflect Polysemy. In Proceedings of the Ninth Joint Conference on Lexical and Computational Semantics, pages 103–113; <https://arxiv.org/abs/2011.09413>.

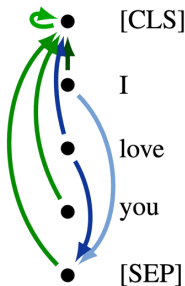
Filtered Graphs from Attention Maps



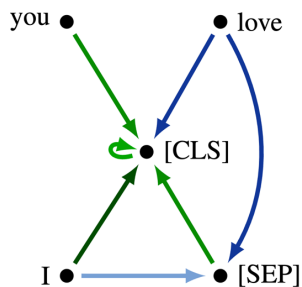
(a)



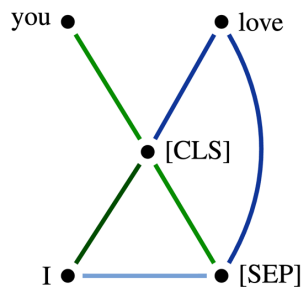
(b)



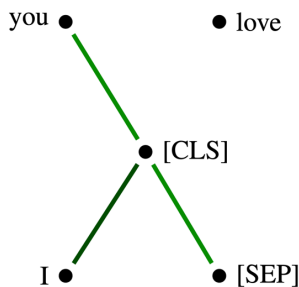
(c)



(d)



(e)

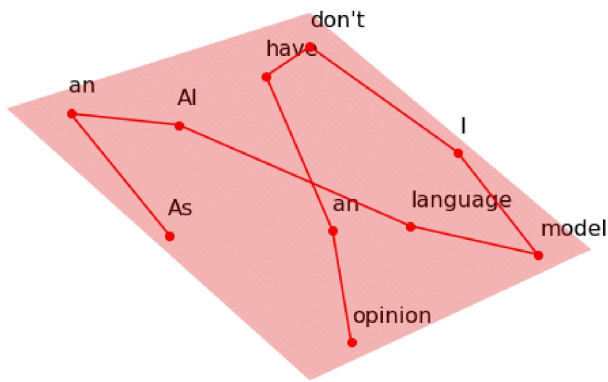


(f)

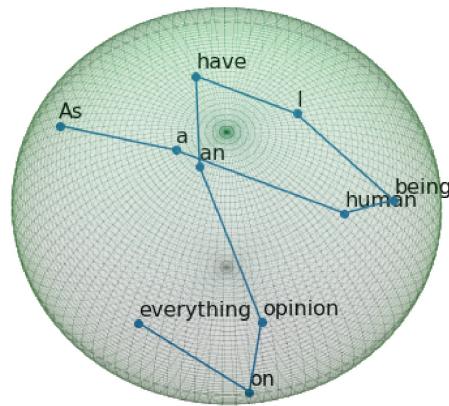
- **Topological features** of attention graphs
→ inputs for a text **classifier**
- Used to **detect**:
AI-written text,
grammatical acceptability,
hallucinations, ...

Laida Kushnareva, Daniil Cherniavskii, Vladislav Mikhailov, Ekaterina Artemova, Serguei Barannikov, Alexander Bernstein, Irina Piontkovskaya, Dmitri Piontkovski, and Evgeny Burnaev. 2021. Artificial Text Detection via Examining the Topology of Attention Maps. *In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 635–649; [arXiv:2109.04825](https://arxiv.org/abs/2109.04825).

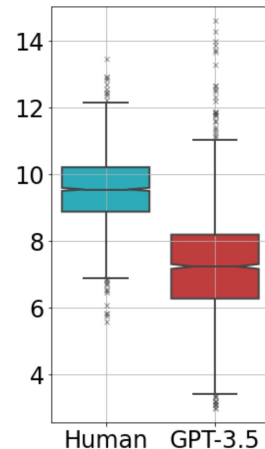
Artificial Text Is Lower-Dimensional Than Human-Written



(a) AI generated



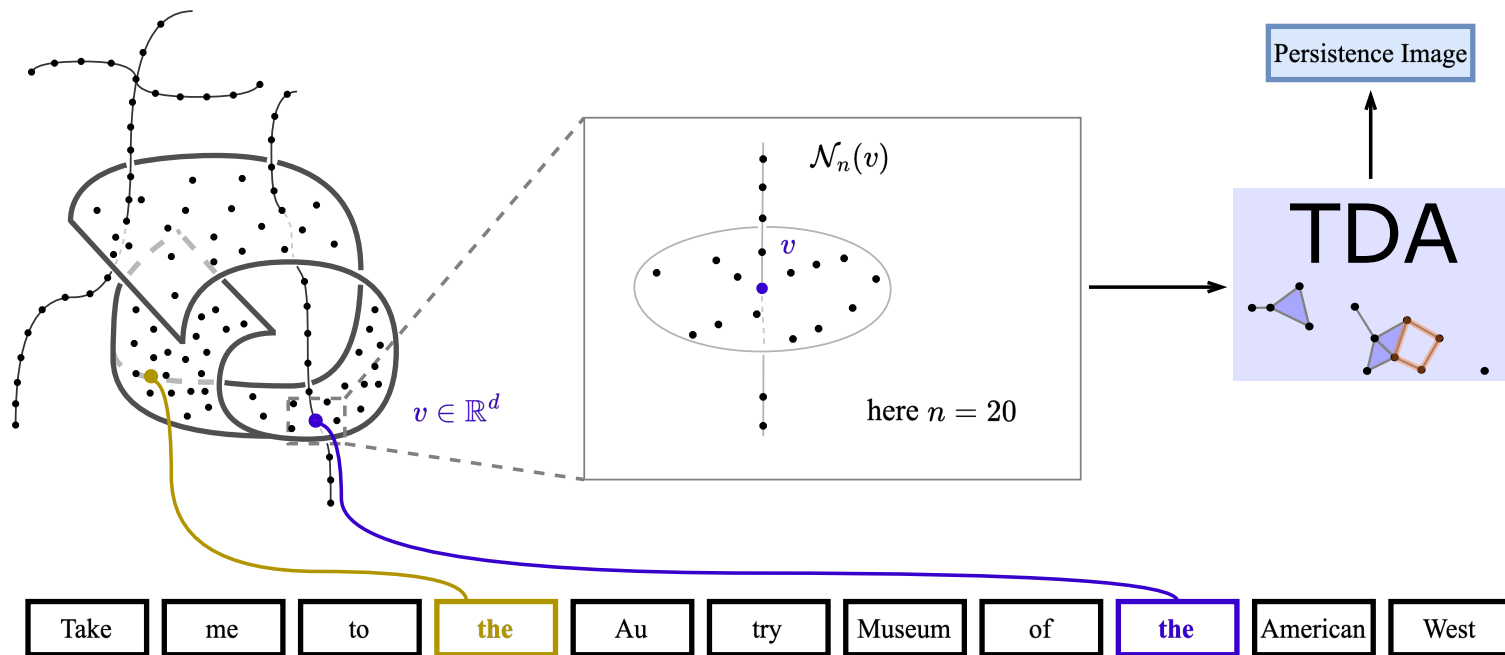
(b) human written



(c)

Tulchinskii, E., Kuznetsov, K., Kushnareva, L., Cherniavskii, D., Nikolenko, S., Burnaev, E., ... & Piontkovskaya, I. (2023). Intrinsic Dimension Estimation for Robust Detection of AI-Generated Texts. *Advances in Neural Information Processing Systems*, 36, 39257-39276; [arXiv:2306.04723](https://arxiv.org/abs/2306.04723).

Augmenting Model Inputs with Topological Features



Benjamin Ruppik, Michael Heck, Carel van Niekirk, Renato Vukovic, Hsien-chin Lin, Shutong Feng, Marcus Zibrowius, and Milica Gasic. 2024. Local Topology Measures of Contextual Language Model Latent Spaces with Applications to Dialogue Term Extraction. *In Proceedings of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 344–356; [arXiv:2408.03706](https://arxiv.org/abs/2408.03706).

Problem

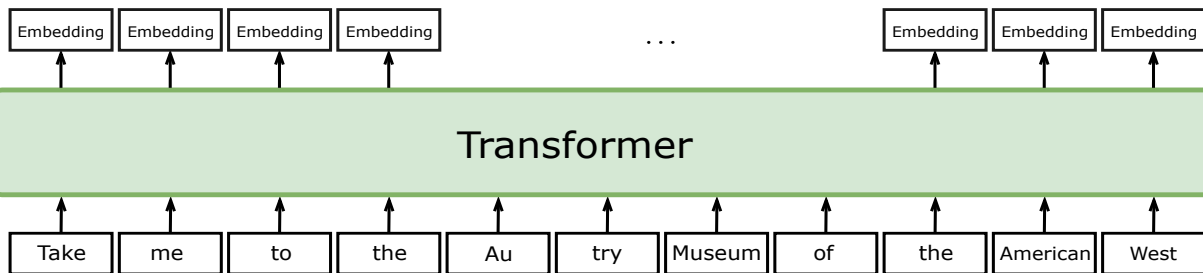
Diagnostics

Supervised &
Task-Specific

Loss/Perplexity

Downstream Task
Performance

...

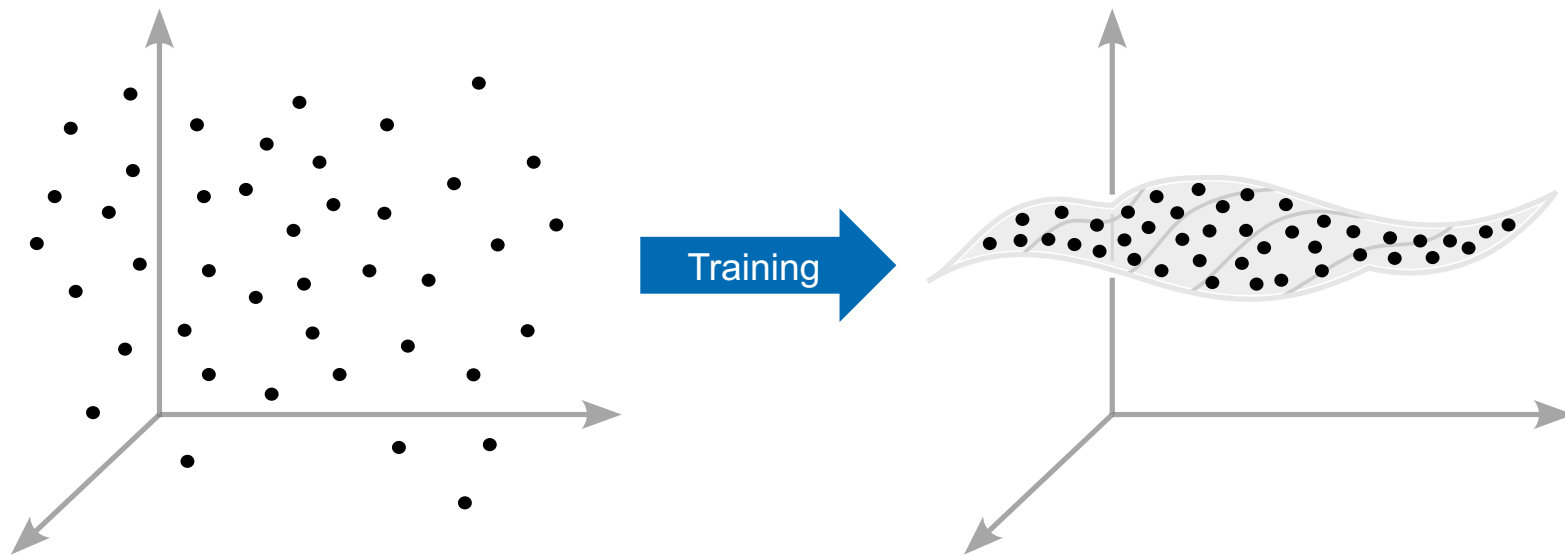


Most useful

Performance on held-out test data

But: Expensive

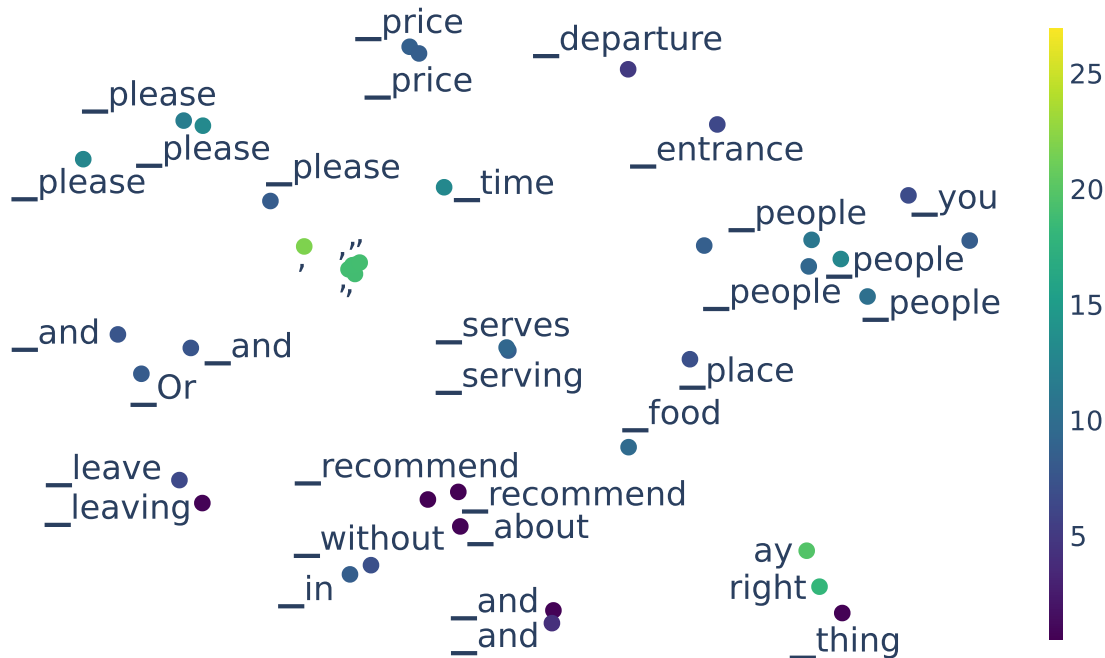
Can Embedding Geometry Reveal Learning Dynamics?



Structural **changes in the embedding space**

→ Unsupervised insights into model behaviour across language tasks?

Method: Local Intrinsic Dimension via TwoNN



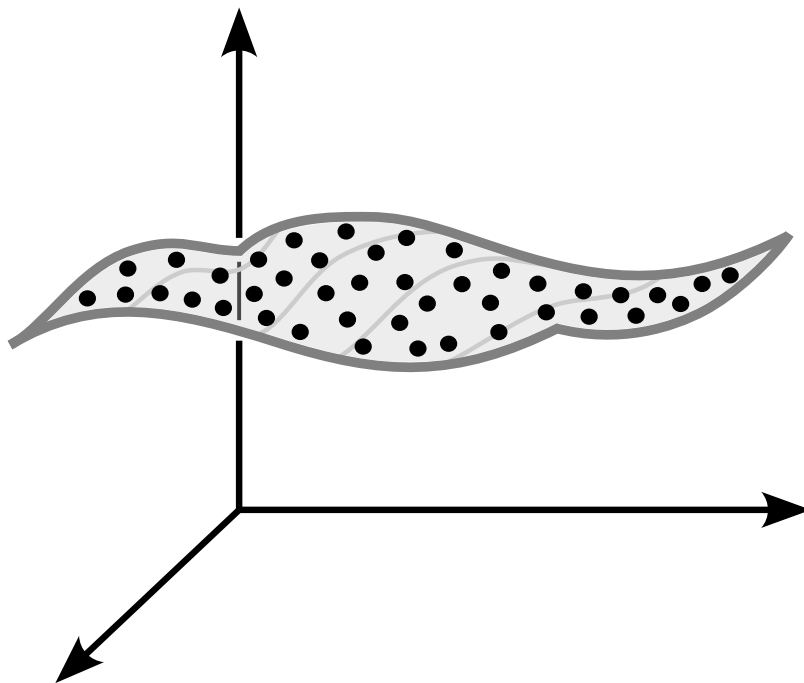
Localized TwoNN estimator (Facco et al., 2017)

Applied to **subsample** of contextual token embeddings

Intrinsic Dimension & How to measure it

Intrinsic dimension (ID)

“Number of relevant coordinates needed to describe a data-generating process.”



COIL-20 dataset



(a) Example images of twenty categories



(b) Part example images of one category

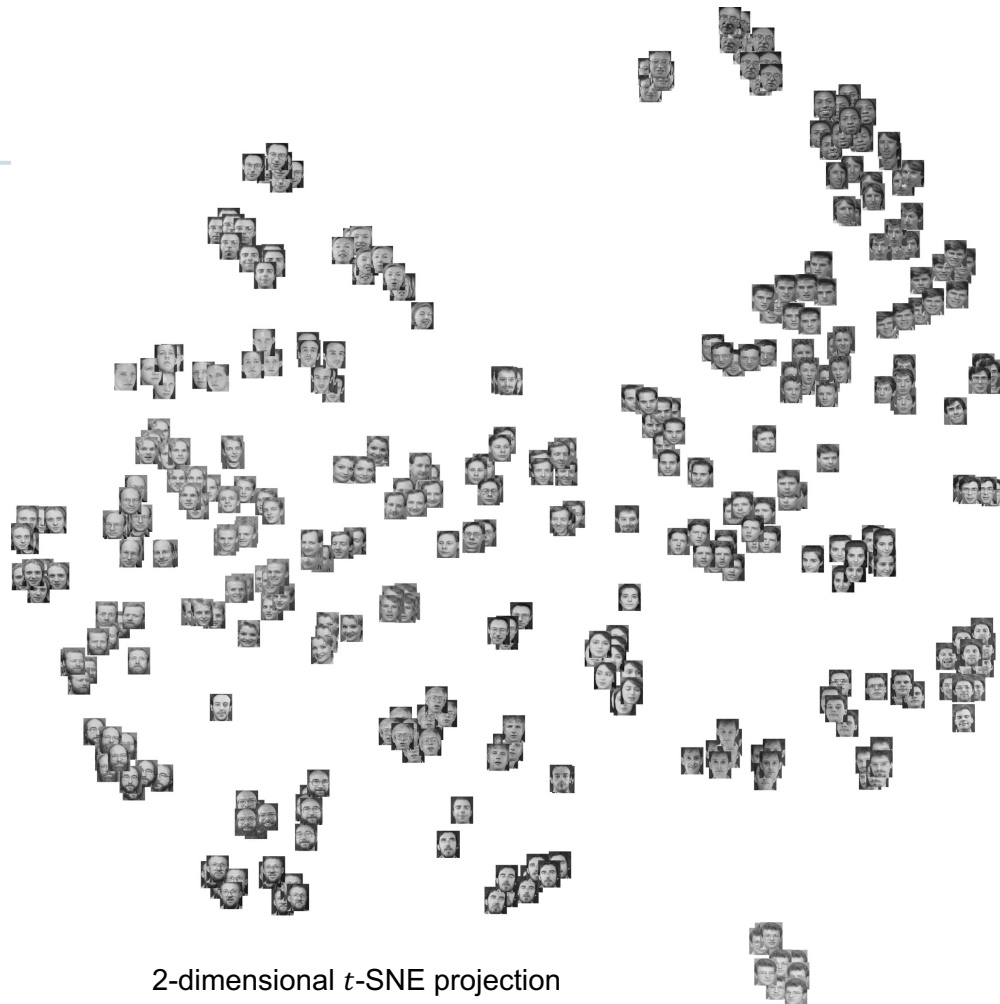
- 128×128 grayscale images
- Twenty objects with 72 poses each



2-dimensional t -SNE projection

Olivetti faces dataset

- 64×64 grayscale images
- 40 subjects with 10 images each



2-dimensional t -SNE projection

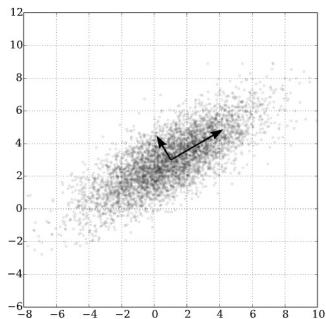
Features of CNN trained on ImageNet

- 50,000 ILSVRC 2012 validation images
- 4,096-dimensional fc7 CNN features

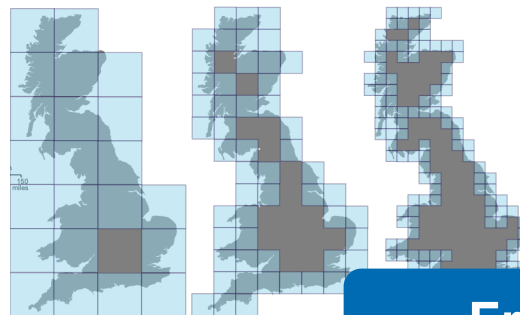


2-dimensional t -SNE projection

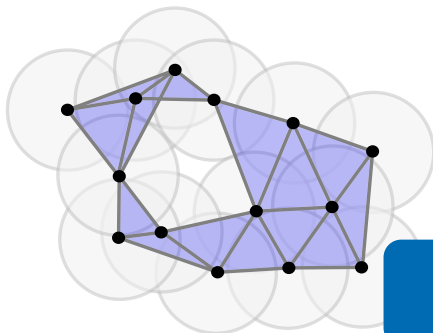
Methods for Intrinsic Dimension (ID) Estimation



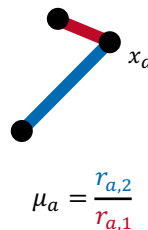
**Projection
Methods**



Fractal



Graph-based



A diagram showing a point x_a connected to two other points by lines of different colors (blue and red). The red line is shorter than the blue line, representing the distance to the nearest neighbor.

$$\mu_a = \frac{r_{a,2}}{r_{a,1}}$$

**Nearest Neighbor
Distance-based**

Poisson Point Processes

Stochastic process generating points randomly located on a subspace

Essential feature

Points occur **independently** of one another.

- **Number of points** in any given finite region follows a **Poisson distribution**
- **Homogeneous & Non-homogeneous** variants



Chewing gum on a sidewalk. The number of pieces on a single tile is approximately Poisson distributed.

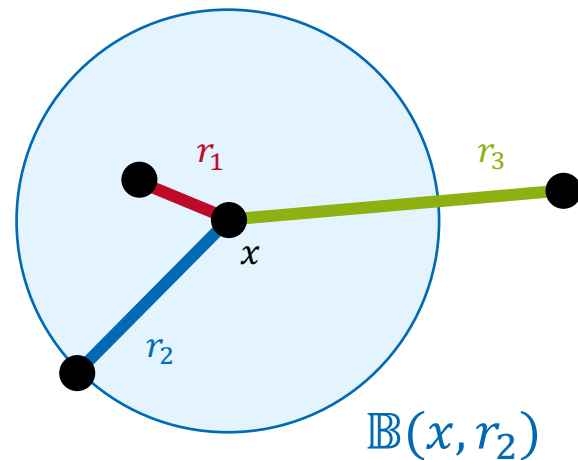
TwoNN Dimension Estimator

Assumption: Data comes from a Poisson point process **supported on a manifold of dimension d** with **constant density ρ** .

→ Points close to each other are uniformly drawn from d -dimensional balls

→ Proportion of points of a given sample which fall into the ball $\mathbb{B}(x, r)$ of radius r is approximately: density $\rho(x)$ times the **volume of the ball**

$$\text{Vol}(\mathbb{B}(x, r)) = \omega_d r^d$$



Upshot

If density is constant → can estimate ID using only the average distances from a point's k NNs.

TwoNN Dimension Estimator

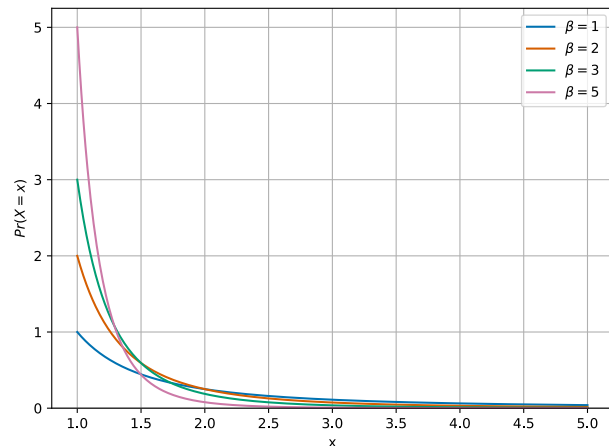
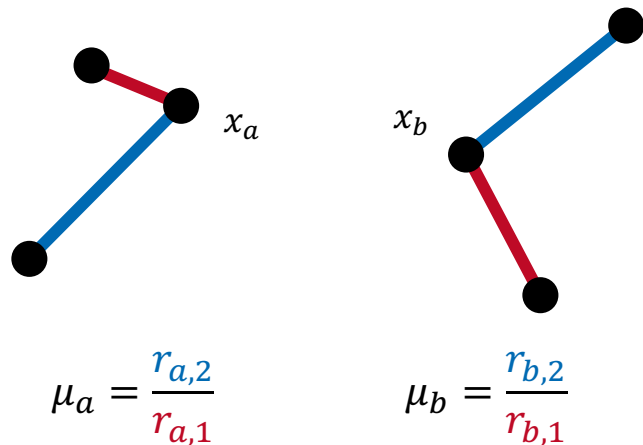
Assumption: Data comes from a Poisson point process **supported on a manifold of dimension d** with **constant density ρ** .

Theorem [Facco et al., 2017]

Ratio of distances $\mu_i = \frac{r_{i,2}}{r_{i,1}}$ between the second and first nearest neighbor of points is **Pareto-distributed** $\text{Pareto}(\alpha, \beta)$ with:

- Scale parameter $\alpha = 1$
- Shape parameter $\beta = d$ the **dimension**.

→ Estimate ID d by **finding shape parameter**



Sequence and Token Subsampling

Problems

- **Large datasets** make computing local intrinsic dimensions **for every point** infeasible
- If datasets have drastically **different number of tokens** → their embedding spaces are not comparable

Example

MultiWOZ 2.1 training split contains:

- 8,438 multi-domain task-oriented dialogues
- with 113,556 utterances
- RoBERTa-tokenizer: Results in **2,165,304 tokens**

Sequence and Token Subsampling

Solution

Subsampling!

_I _would _like _a _tax i _from _Saint _John ' s _college _to _P izza _H ut _Fen _Dit ton .

_What _time _do _you _want _to _leave _and _what _time _do _you _want _to _arrive _by ?

_I _want _to _leave _after _ 1 7 : 1 5 .

_Book ing _completed ! _your _tax i _will _be _blue _h onda _Contact _number _is _ 0 7 2 1 8 0 6 8 5 4 0 .

_Thank _you _for _all _the _help ! _I _appreciate _it .

**Parameter
choices**

- M : Sequence subsample size
- N : Token subsample size

Sequence and Token Subsampling

Solution

Subsampling!

_I _would _like _a _tax i _from _Saint _John ' s _college _to _P izza _H ut _Fen _Dit ton .

_What _time _do _you _want _to _leave _and _what _time _do _you _want _to _arrive _by ?

_I _want _to _leave _after _ 1 7 : 1 5 .

_Book ing _completed ! _your _tax i _will _be _blue _h onda _Contact _number _is _ 0 7 2 1 8 0 6 8 5 4 0 .

_Thank _you _for _all _the _help ! _I _appreciate _it .

**Parameter
choices**

- M : Sequence subsample size
- N : Token subsample size

Sequence and Token Subsampling

Solution

Subsampling!

_I _would _like _a _tax i _from _Saint _John ' s _college _to _P izza _H ut _Fen _Dit ton .

_What _time _do _you _want _to _leave _and _what _time _do _you _want _to _arrive _by ?

_I _want _to _leave _after _ 1 7 : 1 5 .

_Book ing _completed ! _your _tax i _will _be _blue _h onda _Contact _number _is _ 0 7 2 1 8 0 6 8 5 4 0 .

_Thank _you _for _all _the _help ! _I _appreciate _it .

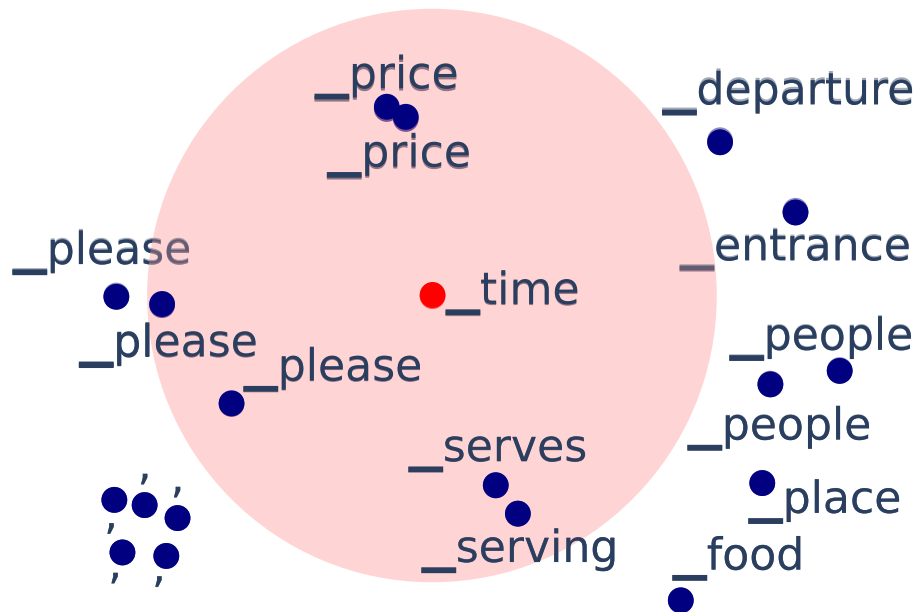
**Parameter
choices**

- M : Sequence subsample size
- N : Token subsample size

Locality of Estimates

Problem

TwoNN estimator relies on local homogeneity assumption

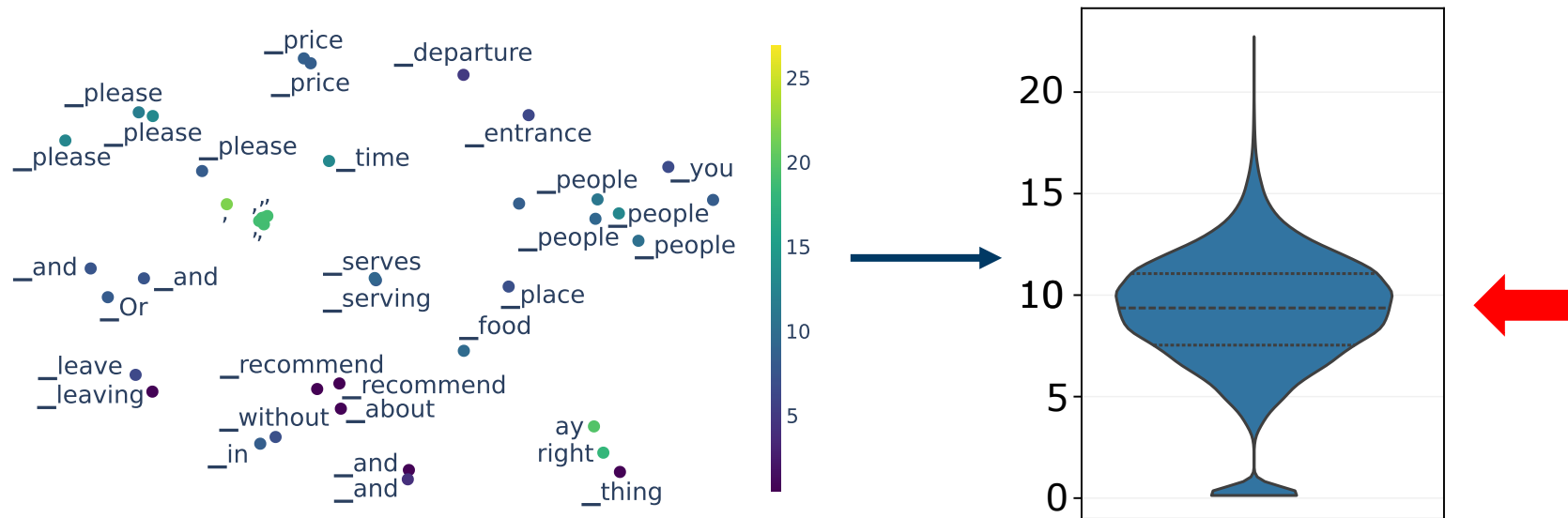


Solution

Compute dimension based on a **neighborhood** (of cardinality L)

Dimensional “Signatures”

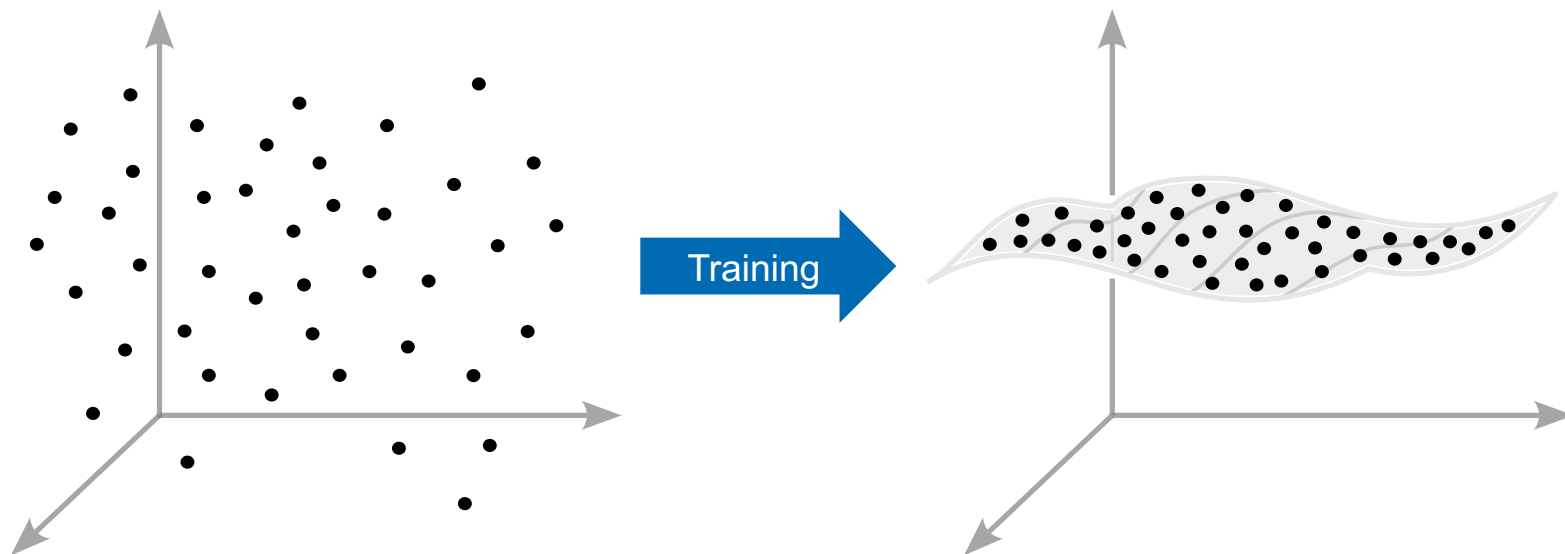
Local variations form **geometric fingerprint**



Mean local dimension

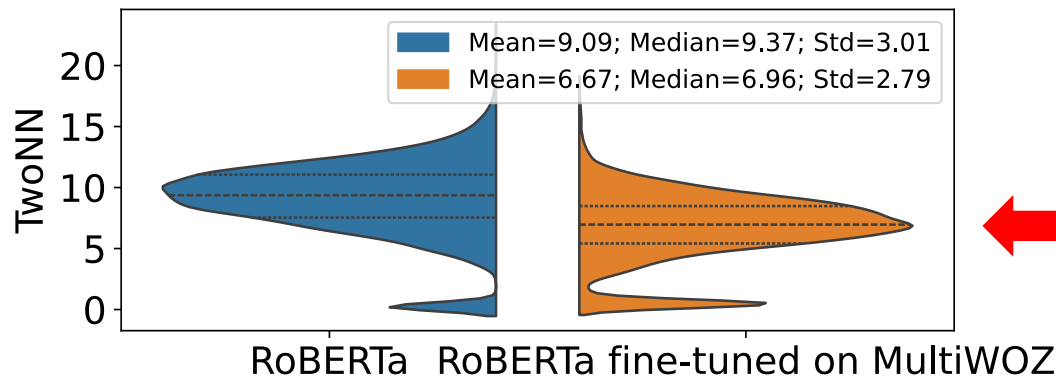
Stable estimate over different data splits and models

Can Embedding Geometry Reveal Learning Dynamics?

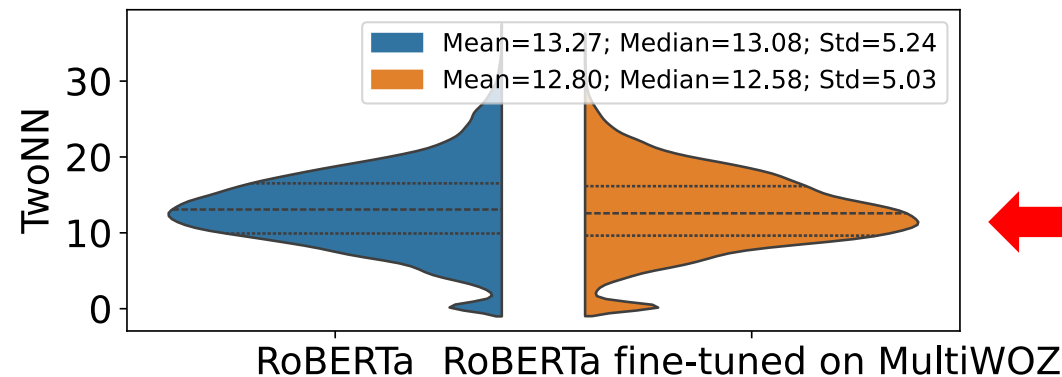


Fine-Tuning Induces Dataset-Specific Dimensional Shifts

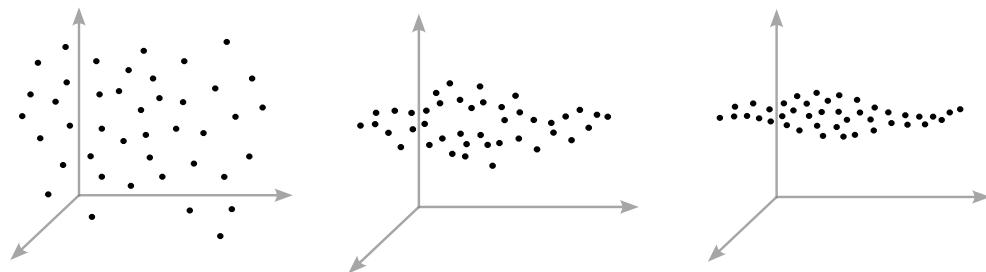
Fine-tuning data



Unrelated data



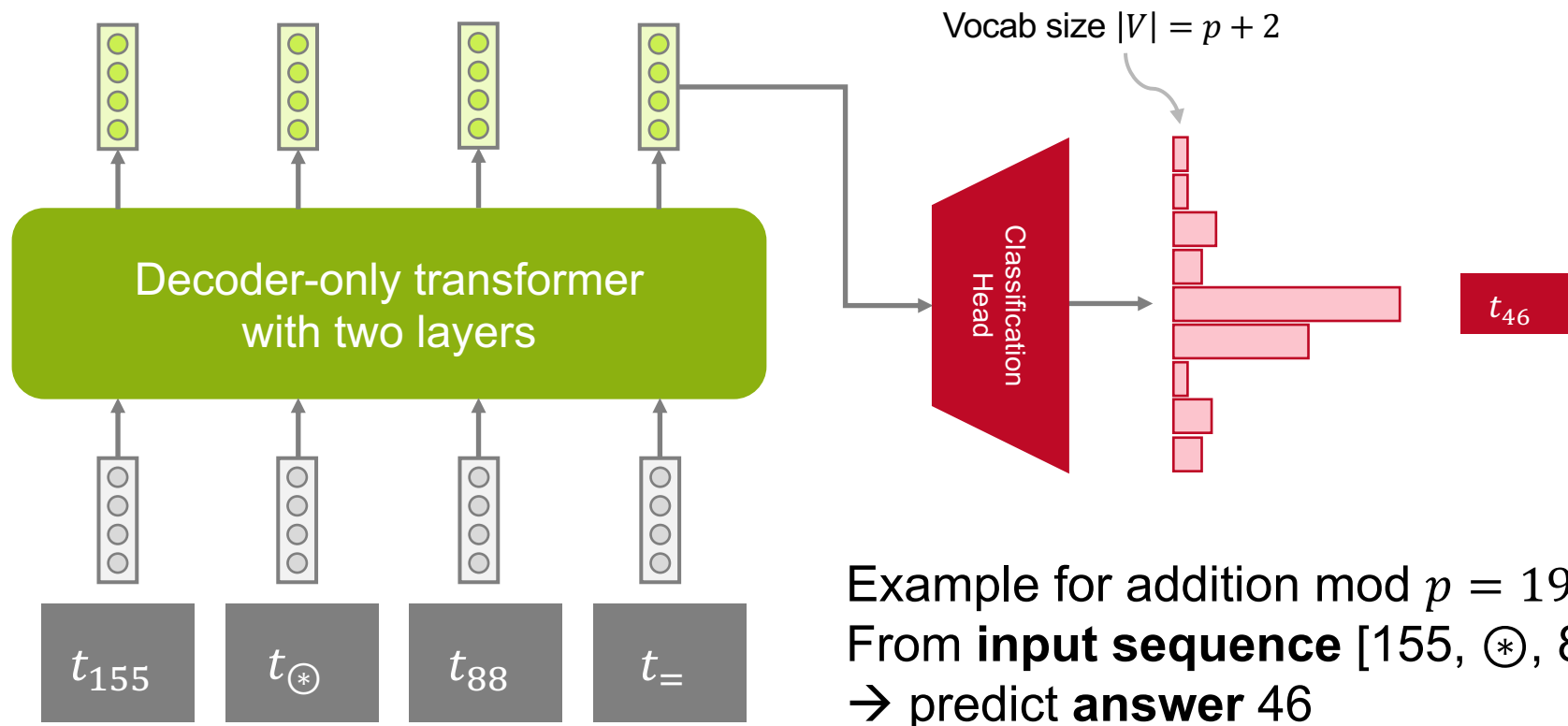
Monitor Local Intrinsic Dimension at Every Step of Training



We identify distinct qualitative trends

- Grokking
- Convergence
- Overfitting

Artificial Arithmetic Task



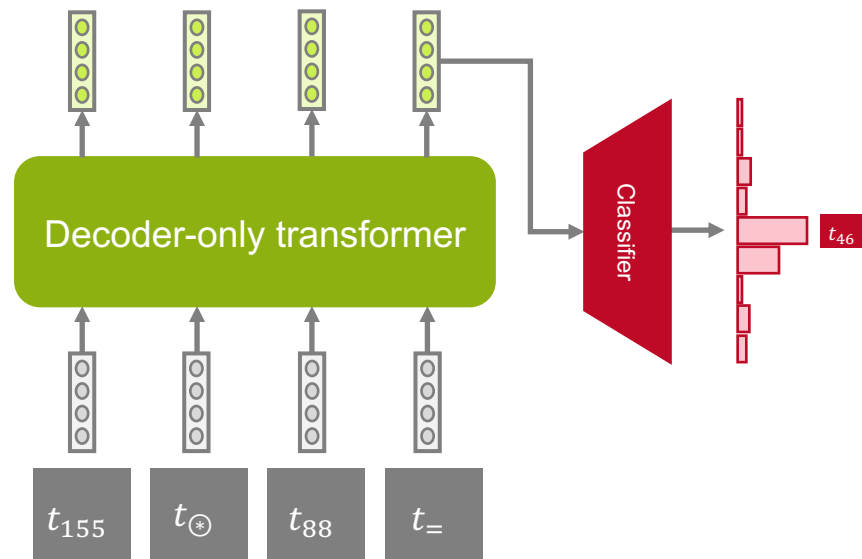
Artificial Arithmetic Task

Addition mod p

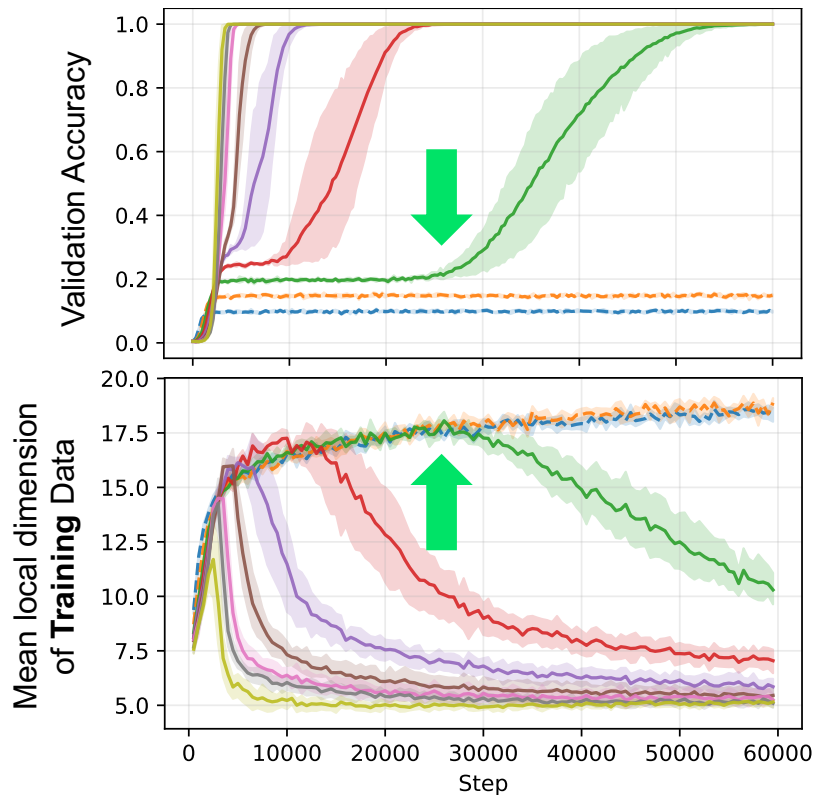
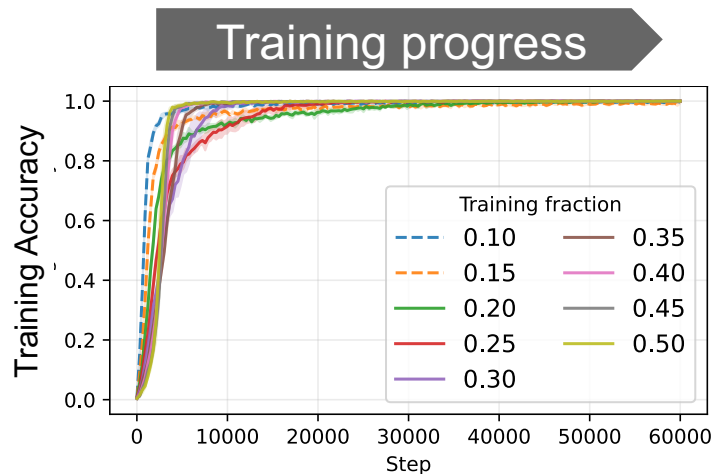
→ p^2 well-formed sequences $[a, \odot, b, =, c]$

Random subset of 10% to 50% of sequences for **training**

Validate on remaining sequences by measuring prediction **accuracy**



Dimension Drop Anticipates **Grokking**



Downstream Task: Dialogue State Tracking (DST)

Task: Maintain a structured representation of the dialogue
→ cumulative memory of **user goal**

U: i'm looking for an **expensive** restaurant in the **center** of town.

S: there are 33 restaurants [...]. would you like to narrow your search by type of food?

U: i **do not care** about food. surprise me.

S: **fitzbillies restaurant** serves british food, [...]. would you like to hear about any others? [...].

U: that sounds fine. can you book it please and get me the reference number?

S: sure , what day and time would you like and how many people ?

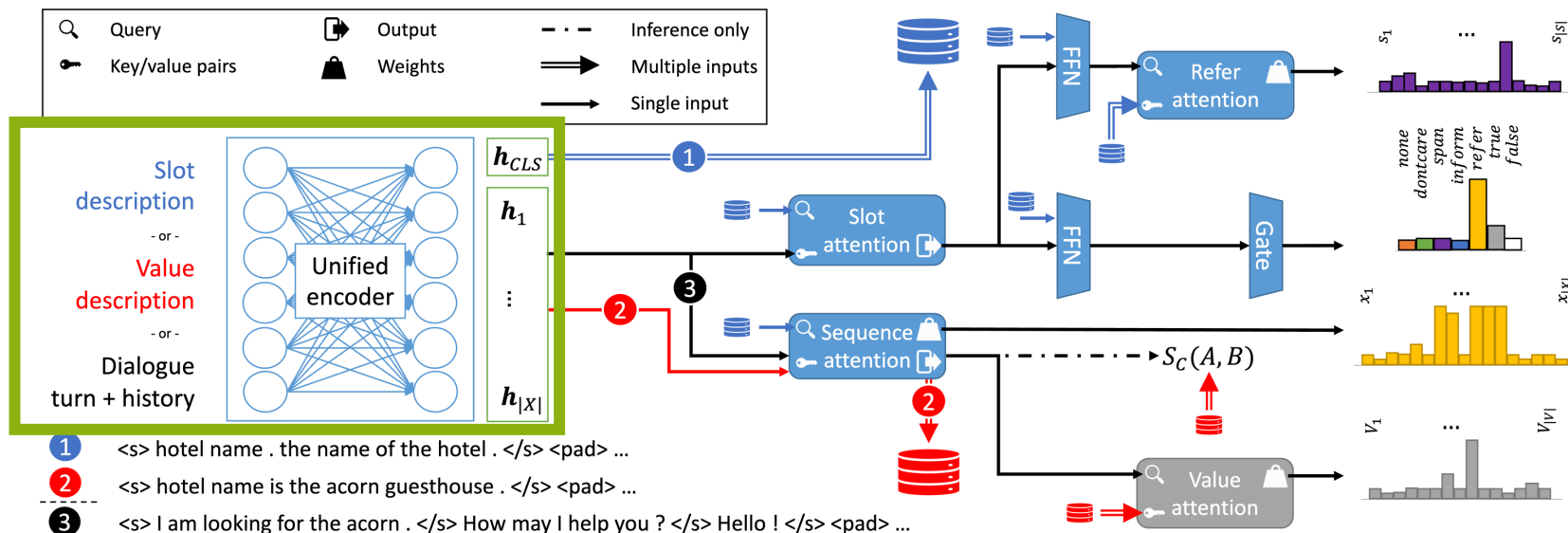
U: i would like a table for **5** at **11:30** on **tuesday** [...]

S: okay, the booking was successful. [...]. is there anything else i can help you with?

U: i'm also looking for a place to stay. it needs [...] **free wifi** and [be] in the **same area as the restaurant.**

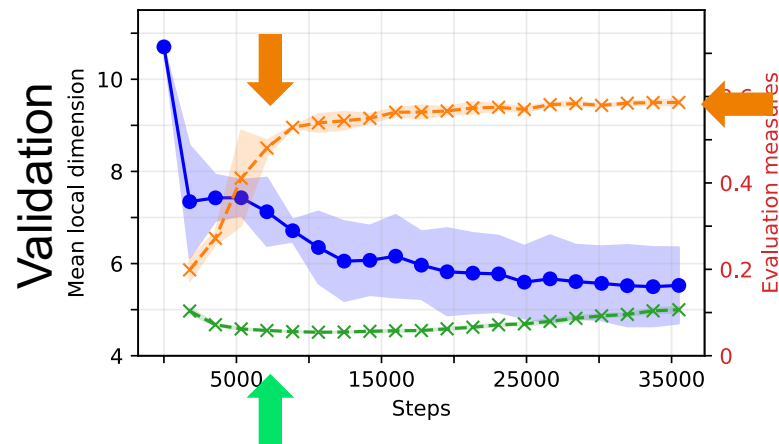
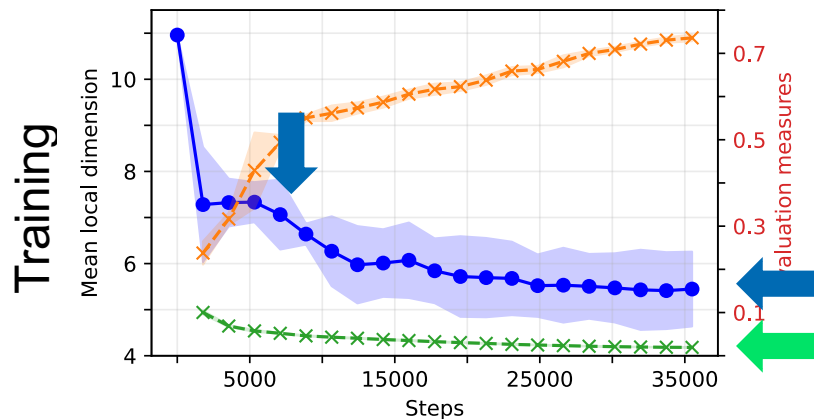
Michael Heck, Carel van Niekerk, Nurul Lubis, Christian Geishausen, Hsien-Chin Lin, Marco Moresi, and Milica Gasic. 2020. TripPy: A Triple Copy Strategy for Value Independent Neural Dialog State Tracking. *In Proceedings of the 21th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 35–44; [arXiv:2005.02877](https://arxiv.org/abs/2005.02877).

TripPy-R: Sequence-Tagging Based DST model



Michael Heck, Nurul Lubis, Carel van Niekerk, Shutong Feng, Christian Geishausen, Hsien-Chin Lin, and Milica Gašić. 2022. Robust Dialogue State Tracking with Weak Supervision and Sparse Data. In: *Transactions of the Association for Computational Linguistics*, 10:1175–1192; [arXiv:2202.03354](https://arxiv.org/abs/2202.03354).

Dimension Stabilization Mirrors Learning Convergence



- Mean local dim.
- x- Joint Goal Accuracy
- x- Loss

Downstream Task: Emotion Recognition (ERC)

Task: Classifying dialogue utterances into one of seven emotion classes

"Yes, I've heard great things about that."



Excited

"Sorry, I misremembered the date. I actually want to check into one room on Saturday this week."



Apologetic

"You are horrible and not helpful at all!"



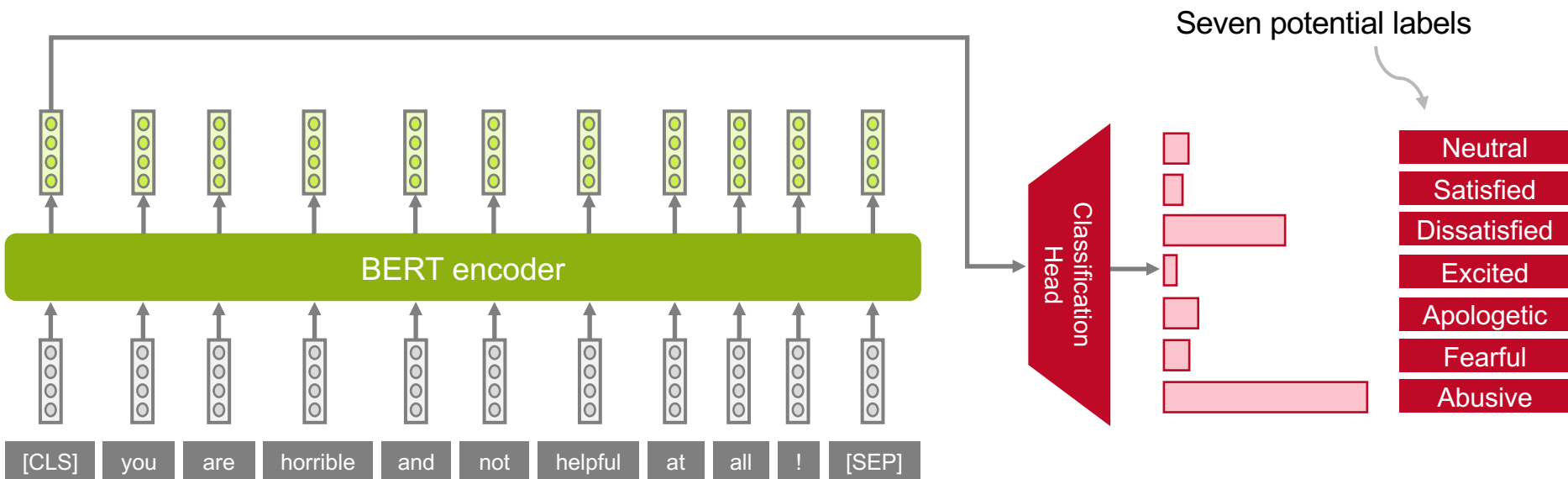
Abusive

**EmoWOZ
dataset**

User emotion annotations for all dialogues from MultiWOZ (plus additional human-machine dialogues)

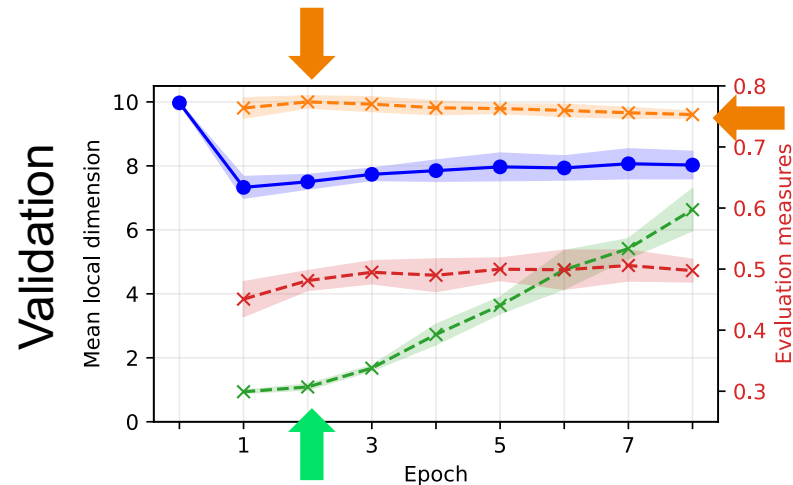
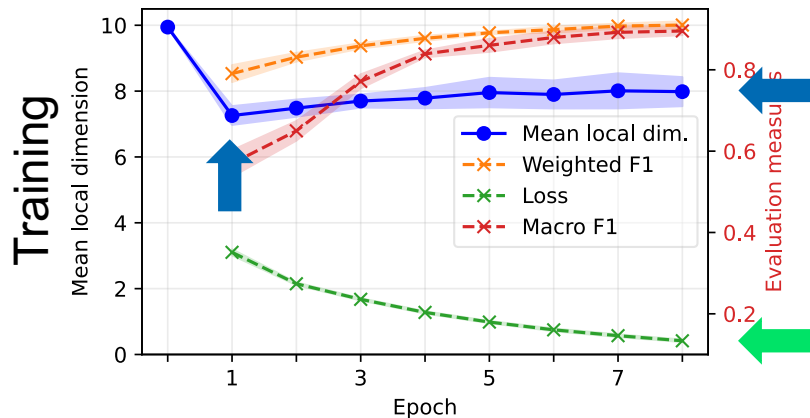
Shutong Feng, Nurul Lubis, Christian Geishauser, Hsien-chin Lin, Michael Heck, Carel van Niekerk, and Milica Gasic. 2022. EmoWOZ: A Large-Scale Corpus and Labelling Scheme for Emotion Recognition in Task-Oriented Dialogue Systems. *In Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 4096–4113; [arXiv:2109.04919](https://arxiv.org/abs/2109.04919).

ERToD model: Classifier on Top of Feature Extractor



Shutong Feng, Nurul Lubis, Benjamin Ruppik, Christian Geischauser, Michael Heck, Hsien-chin Lin, Carel van Niekerk, Renato Vukovic, and Milica Gasic. 2023. From Chatter to Matter: Addressing Critical Steps of Emotion Recognition Learning in Task-oriented Dialogue. In *Proceedings of the 24th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 85–103; [arXiv:2308.12648](https://arxiv.org/abs/2308.12648).

Dimension Increase Detects Overfitting



Initial drop followed by a rise in dimension $\leftrightarrow$ Overfitting

Across diverse tasks, a sustained **drop in mean local dimension** reliably predicts **improved generalization**.

Less is More: Local Intrinsic Dimensions of Contextual Language Models

Benjamin Matthias Ruppik, Julius von Rohrscheidt, Carel van Niekerk, Michael Heck,
Renato Vukovic, Shutong Feng, Hsien-chin Lin, Nurul Lubis,
Bastian Rieck, Marcus Zibrowius, Milica Gašić

Dialog Systems and Machine Learning Group, Faculty of Mathematics
and Natural Sciences, **Heinrich Heine University Düsseldorf**, Germany
Institute of AI for Health, **Helmholtz Munich**, Germany
Technical University of Munich, Germany
AIDOS Lab, **University of Fribourg**, Switzerland

To appear at NeurIPS 2025



Heinrich Heine
Universität Düsseldorf



UNIVERSITÉ DE FRIBOURG
UNIVERSITÄT FREIBURG



HELMHOLTZ MUNICH



European Research Council
Established by the European Commission

Ministerium für
Kultur und Wissenschaft
des Landes Nordrhein-Westfalen



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra



Google Cloud



Heinrich Heine
Universität Düsseldorf



We are looking forward to your questions!

Our Paper and Code

- Benjamin Ruppik, Julius von Rohrscheidt, Carel van Niekerk, Michael Heck, Renato Vukovic, Shutong Feng, Hsien-chin Lin, Nurul Lubis, Bastian Rieck, Marcus Zibrowius, Milica Gašić (2025).
Less is More: Local Intrinsic Dimensions of Contextual Language Models.
In: The Thirty-ninth Annual Conference on Neural Information Processing Systems, NeurIPS 2025, San Diego, USA.
- [arXiv:2506.01034](https://arxiv.org/abs/2506.01034)
- https://github.com/aidos-lab/Topo_LLM_public
- <https://github.com/aidos-lab/grokking-via-lid>

References

- Facco, E., d'Errico, M., Rodriguez, A. et al. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Sci Rep* 7, 12140 (2017); [doi:10.1038/s41598-017-11873-y](https://doi.org/10.1038/s41598-017-11873-y); [arXiv:1803.06992](https://arxiv.org/abs/1803.06992).
- Denti, F., Doimo, D., Laio, A., & Mira, A. (2021). Distributional Results for Model-Based Intrinsic Dimension Estimators; [arXiv:2104.13832](https://arxiv.org/abs/2104.13832).
- Power, A., Burda, Y., Edwards, H., Babuschkin, I., & Misra, V. (2022). Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets; [arXiv:2201.02177](https://arxiv.org/abs/2201.02177).